Cassandra Installation

1.Install WSL on your pc if you're using windows (for versions of cassandra [4.xx.xx](4.xx.xx)+)

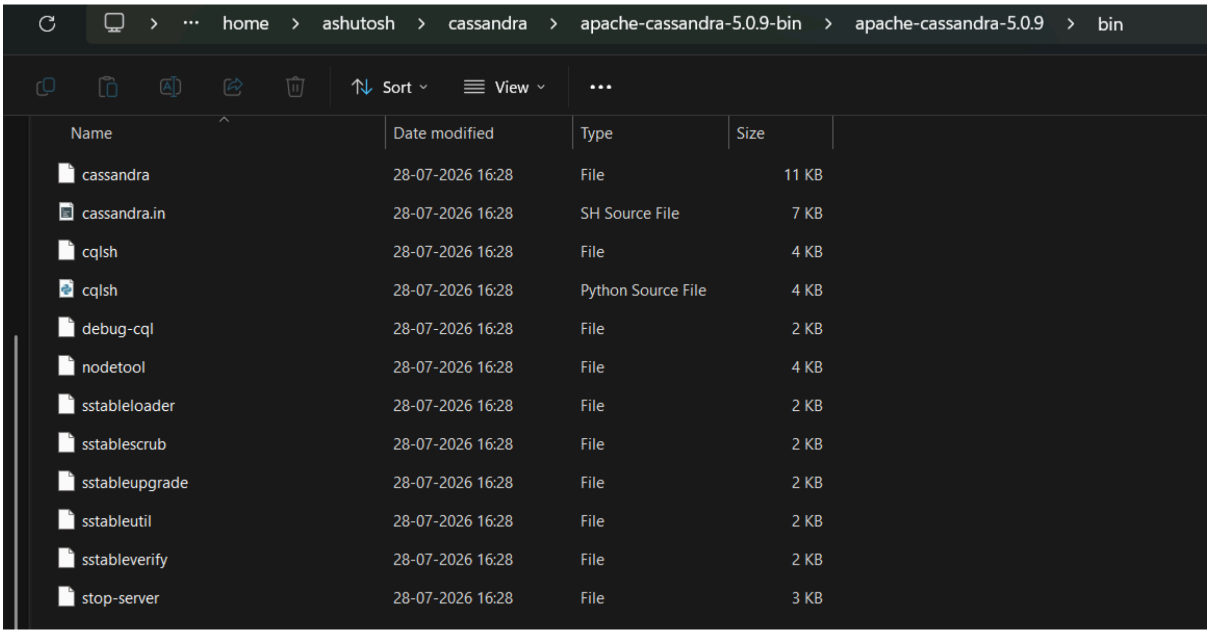
2.configure WSL and install ubuntu resolute.
Update ubuntu using:
sudo apt update
sudo apt upgrade -y

3.Download Cassandra ([5.xx.xx](5.xx.xx) stable) from:
https://cassandra.apache.org/_/download.html

4. If you're on windows then move the install (and unzipped cassandra.tar.gz file) to \ubuntu\home\<user>\cassandra
The path should look like \\wsl.localhost\Ubuntu\home\ashutosh\cassandra\

5. In cassandra\bin folder, open a command prompt/powershell and run WSL to initiate wsl environment inside the folder.

6. You should see files such as cassandra and cqlsh, the first one is cassandra server, the second one is cql (cassandra query language) shell which runs the queries. The file structure should look like this:-



7. If there are any error then copy paste the error codes or texts to chatgpt or gemini. Common errors are due to unavailability of java and python (needs java jdk 17+, download using "sudo apt install -y openjdk-17-jdk")

** if python is not available or beyond 3.13 then do the following:-

(ALL COMMANDS INSIDE WSL INTERFACE)
In order of execution:-
a)sudo apt install -y \
    build-essential \
    curl \
    git \
    libssl-dev \
    zlib1g-dev \
    libbz2-dev \
    libreadline-dev \
    libsqlite3-dev \
    libncurses-dev \
    xz-utils \
    tk-dev \
    libffi-dev \
    liblzma-dev \
    libxml2-dev \
    libxmlsec1-dev \
    libgdbm-dev \
    libnss3-dev
b)curl https://pyenv.run | bash
c)echo 'export PYENV_ROOT="$HOME/.pyenv"' >> ~/.bashrc
echo '[[-d $PYENV_ROOT/bin]] && export PATH="$PYENV_ROOT/bin:$PATH"' >> ~/.bashrc
echo 'eval "$(pyenv init - bash)"' >> ~/.bashrc
d)source ~/.bashrc
e)pyenv --version #this should show something like 2.xx.xx
f)pyenv install 3.13
g)pyenv global 3.13

8. Then open a command prompt inside the bin folder as seen in the picture above, Run cassandra and then keep that terminal open, wait and wait until you see something like this:

Node localhost/127.0.0.1:7000 state jump to NORMAL

Then open another terminal while keeping the first one OPEN, do not close it as its running the server for cassandra
In the 2nd command prompt, run cqlsh and wait until it shows something like this:

Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.9 | CQL spec 3.4.7 | Native protocol v5]

9. Cassandra installation and server setup has been completed. Keep the first terminal in which you ran cassandra open and run your queries on the second terminal that ran cqlsh which is the shell for cassandra.